

Ordinary Differential Equation

1st Year Even Semester

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Topic covered throughout this course:

- ✓ Formation of ODE
- ✓ First order ODE:
 - Separable Equation
 - Homogeneous Function
 - Integrating Factor
 - Exact Equation
 - Bernoulli's Equation
 - Application of first order
- ✓ Second order ODE
 - Homogeneous with constant coefficient
 - non-homogeneous ODE (with constant co-efficient)
 - Application of 2nd order ODE
- ✓ Series Solution:
 - a) Legendre equation
 - i. Solution
 - ii. Orthogonal Property
 - iii. Generative Function and it's property
 - b) Bessel Equation

Detailed Topic and Problems:

1. radioactive decay: Solve this equation, $\frac{dN}{dt} = -\lambda N$
2. Integrating factor:
3. Solve this equation: $dQ = C_v dT + p dV$ and prove that if the entropy is zero,
Then prove that $p v^\gamma = k$. [$S = C_v \ln T + R \ln T$]
4. Solve : i) $y dx - x dy = 0$ ii) $y dx + x dy = 0$.
5. Exact equation:
 $M dx + N dy = 0$;
6. Solve:
 - a. $(x+y) dx + x dy = 0$

b. $(x^2+y)dx + (x-y^2)dy = 0$

c. $(x+y) dx + ydy = 0$

7. Solve: $\frac{di}{dt} + \frac{R}{L}i = \frac{v}{L}$; $i = Ae^{-\frac{R}{L}t} + \frac{v}{L}$

8. Simple Harmonic & Damped Harmonic:

a. $m \frac{d^2x}{dt^2} + \omega^2 x = 0$ b. $\frac{d^2x}{dt^2} + 2c \frac{dx}{dt} + \omega^2 x = 0$

9. Solve: $e^{2y} x dx + e^{2y} (x^2 + y^2 + y) dy = 0$

10. Solve: $m \frac{d^2\theta}{dt^2} + \frac{1}{t} \frac{d\theta}{dt} + \frac{\theta}{t} = 0$

11. Resonance Condition

12. Solve:

a) $\frac{dy}{dx} + y = 0$; $y = Ae^{-x}$

b) $\frac{dy}{dx} + y = e^{-x}$; $y = Ae^{-x} + xe^{-x}$

c) $\frac{dy}{dx} + 2y = e^{-2x}$; $y = Ae^{2x} + xe^{-2x}$

d) $\frac{dy}{dx} + y = e^{-ix}$

e) $\frac{d^2y}{dx^2} + y = \sin x$

f) $(D^2 + RD + 1)y = \sin x$

g) $\frac{dy}{dx} + y = e^{2x}$;

h) $\frac{d^2x}{dt^2} + 2 \frac{dx}{dt} + x = e^{-2t}$

i) $\frac{d^2x}{dt^2} + x = \sin 2t$

j) $\frac{d^2x}{dt^2} + \omega^2 x = \sin vt$; if i) $v = \omega$ ii) $v \neq \omega$

13. Legendre Equation:

a. $(1 - x^2)y'' - 2xy' + \lambda y = 0$